# Yutaka Shimizu

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## Objective

To obtain a challenging position in the field of reinforcement learning, planning, and control for robots.

## Education

**University of California, Berkeley** — *Berkeley, CA*

Ph.D. student in Mechanical Engineering (Reinforcement Learning, Robotics, Motion Planning, Control) — *Aug. 2023 - May. 2027 (Expected)*

- Academic Advisor: Prof. Masayoshi Tomizuka
- Minor: Machine Learning, Optimization
- GPA: 4.00

**The University of Tokyo** — *Tokyo, Japan*

M.E. in Information Science and Technology — *Apr. 2019 - Mar. 2021*

- GPA: 3.84

**Waseda University** — *Tokyo, Japan*

B.S. in Mechanical Engineering — *Apr. 2015 - Mar. 2019*

- GPA: 3.79
- Thesis: "Improving the Accuracy of Mathematical Models for Predicting Sudden Changes in Driver's Condition"

## Research Interest

Autonomous Driving, Robotics, Control, Path/Trajectory Planning, Reinforcement Learning, Offline Reinforcement Learning, Game Theory, Optimization

## Publications and Poster Sessions

1. **Yutaka Shimizu** and Masayoshi Tomizuka "Adaptive Linear Path Model-Based Diffusion" *IEEE/RSJ International Conference on Robotics and Automation (ICRA)*, 2026.
2. **Yutaka Shimizu** and Masayoshi Tomizuka "Bisimulation Metric for Model Predictive Control" *International Conference on Learning Representations (ICLR)*, 2025.
3. **Yutaka Shimizu**, Ashkan Jasour, Maani Ghaffari, Shinpei Kato "Moment-based Kalman Filter: Nonlinear Kalman Filtering with Exact Moment Propagation" *IEEE/RSJ International Conference on Robotics and Automation (ICRA)*, 2023.
4. Hai Zhong, **Yutaka Shimizu**, Jianyu Chen "Chance-Constrained Iterative Linear-Quadratic Stochastic Games" *IEEE Robotics and Automation Letters(RA-L)*.
5. **Yutaka Shimizu**, Takamasa Horibe, Fumiya Watanabe, Shinpei Kato "Jerk Constrained Velocity Planning for an Autonomous Vehicle: Linear Programming Approach." *IEEE/RSJ International Conference on Robotics and Automation (ICRA)*, 2022.
6. Jianyu Chen*, **Yutaka Shimizu***, Wei Zhan, Liting Sun, Masayoshi Tomizuka (Y.Shimizu and J.Chen contributed equally) "Constrained Iterative LQG for Real-time Chance Constrained Gaussian Belief Space Planning." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021. (*contributed equally)
7. **Yutaka Shimizu**, Wei Zhan, Liting Sun, Jianyu Chen, Shinpei Kato, Masayoshi Tomizuka. "Motion Planning for Autonomous Driving with Extended Constrained Iterative LQR." *Dynamic Systems and Control Conference (DSCC)*, 2020.
8. **Yutaka Shimizu**, Takamasa Horibe, Hiroyuki Chishiro, Shinpei Kato. "Path Planning Algorithms for Autoware: Performace Trade-off Details." *Poster Session presented at International Symposium for Advanced Computing and Information Technology (ISACIT)*, 2019.

## Work Experience

**Toyota Research Institute** *Cambridge, MA*
Internship *May. 2026 - Aug. 2026*
- Human Interactive Driving team
- Working on world model and residual RL

**Zoox** *Foster City, CA*
Internship *May. 2025 - Aug. 2025*
- Trajectory generation team
- Working on machine learning project to make the planner more efficient

**University of California, Berkeley** *Berkeley, CA*
Graduate Student Researcher *Aug. 2023 - Present*
- Closed-loop Simulator Research
- Developing a simulator with Python

**TierIV** *Tokyo, Japan*
Planning and Control Researcher *Dec. 2018 - Aug. 2023*
- Researched on the front of optimal control theory, vehicle modeling, trajectory planning, and Robotic Operation System
- Developed Vehicle AStar and QP Velocity Planner with ROS, ROS2 and C++
- Developed Vehicle Steering Controller by Model Vehicle Controller with ROS, ROS2, and C++
- Developed Auto Calibration system with Python
- Contributed for Open Source Software, Autoware

**MSC Lab, University of California, Berkeley** *Berkeley, U.S.A*
Visiting Student(Research Sponsored by JST scholarship) *Feb. 2020 - Feb. 2021*
- Researched on the front of optimal control theory, trajectory planning, and game theory

**PFLab, The University of Tokyo** *Tokyo, Japan*
Graduate Researcher *Apr. 2019 - Mar. 2021*
- Developed Vehicle Continuous-Optimization based Trajectory Planner with ROS and C++

**Fixstars** *Tokyo, Japan*
Internship & Software Engineer *Mar. 2019 - Dec. 2019*
- Designed localization methods(EKF, UKF and etc.) for a mobile robot
- Speed up the Particle Filter algorithm using CUDA for parallel calculation
- Experimented using a real mobile robot to check the algorithm

**Kusaka Laboratory, Waseda University** *Tokyo, Japan*
Undergraduate Researcher *Sep. 2017 - Mar. 2019*
- Modeled the human cardiovascular system using fluid dynamics in MATLAB
- Completed an Independent Project on

**Kyoto Robotics** *Tokyo, Japan*
Spring Internship *Feb. 2018 - Mar. 2018*
- Designed communication protocols between two manipulation robots in C and C++
- Designed Image Process Module for detecting soft objects

## Awards & Certificates

### HONORS & AWARDS

2024 **Japan Student Service Organization Scholarship**, Government-financed scholarship *Japan*
2020 **CREST JST Scholarship**, The University Of Tokyo *Tokyo, Japan*

### Certificates

2020 **TOEFL Score 103**, English Proficiency Test *Tokyo, Japan*
2016 **Level4 HSK**, Chinese Proficiency Test *Tokyo, Japan*